

LabSheet-4

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Roll No. 50

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_and_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-selectedcolumns.csv")

print(df.head())
print(df.shape)
```

	Age	Attrition	BusinessTravel	DailyRate	Department
	41	Yes	Travel_Rarely	1102	Sales

\					
0					
1	49	No	Travel_Frequently	279	Research & Development
2	37	Yes	Travel_Rarely	1373	Research & Development
3	33	No	Travel_Frequently	1392	Research & Development
4	27	No	Travel_Rarely	591	Research & Development

	DistanceFromHome	Education	EducationField	EmployeeCount
EmployeeNumber				
	1	2	Life Sciences	10
				1
	8	1	Life Sciences	11
				2
	2	2	Other	12
				4
	3	4	Life Sciences	13
				5
	2	1	Medical	14
				7

```
(1470, 10)

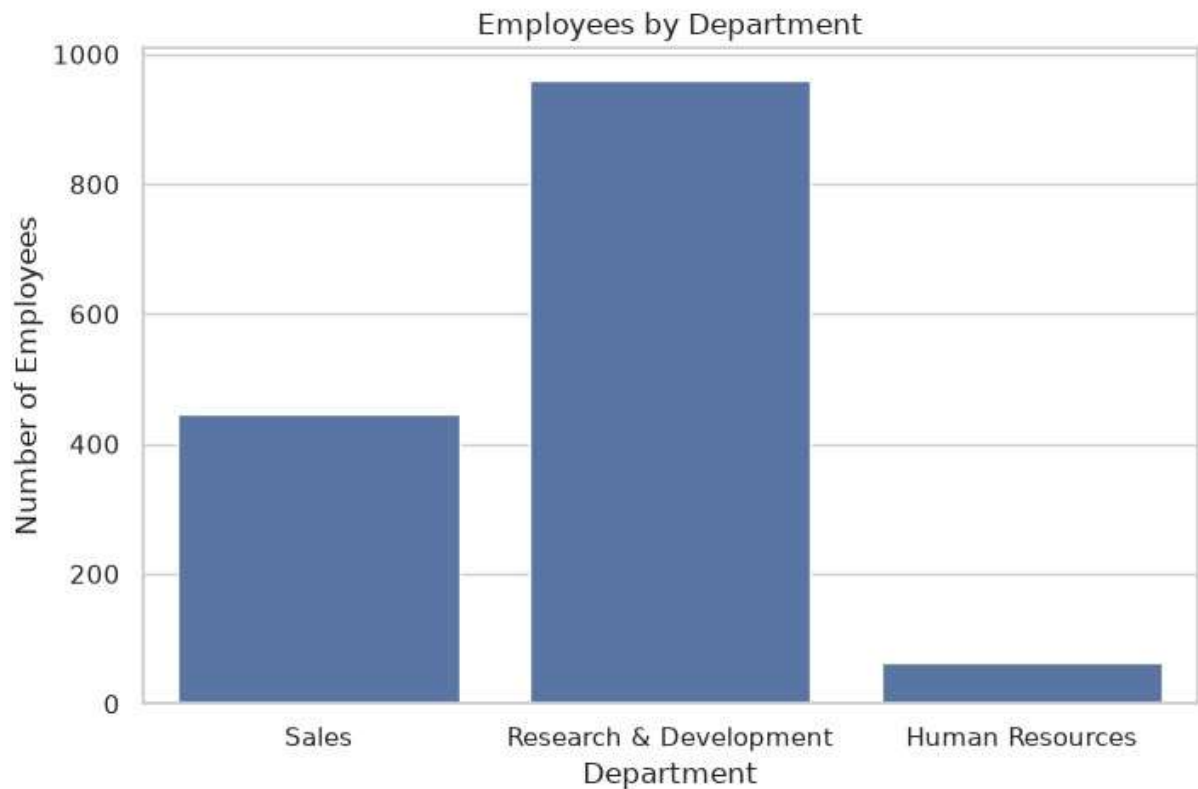
print(df.columns.tolist())

['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
```

```
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',  
'EmployeeNumber']  
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x="Department")
```

```
plt.title("Employees by Department")
plt.xlabel("Department") plt.ylabel("Number of
Employees") plt.show()
```



```
df["AgeGroup"] = pd.cut(
df["Age"],
    bins=[18, 25, 30, 35, 40, 50, 60],
    labels=["18-25", "26-30", "31-35", "36-40", "41-50", "51-60"]
)

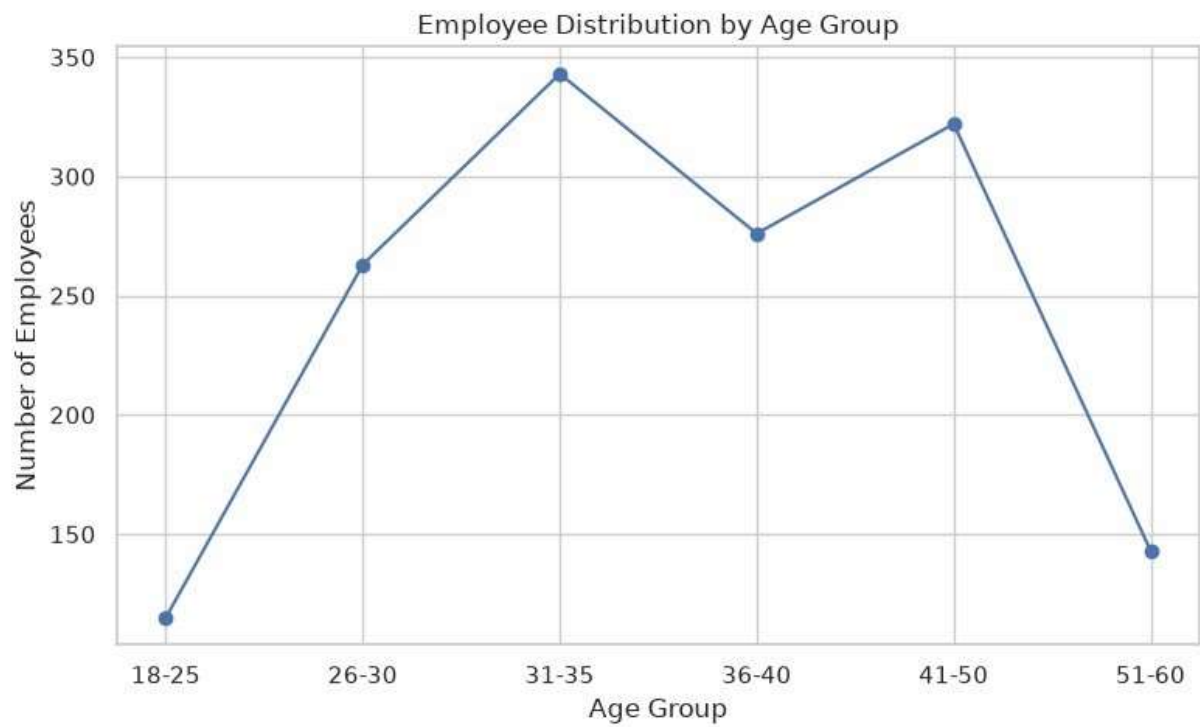
age_count = df["AgeGroup"].value_counts().sort_index()

plt.figure(figsize=(9, 5))

plt.plot(age_count.index, age_count.values, marker="o")

plt.title("Employee Distribution by Age Group")
plt.xlabel("Age Group") plt.ylabel("Number of
Employees")

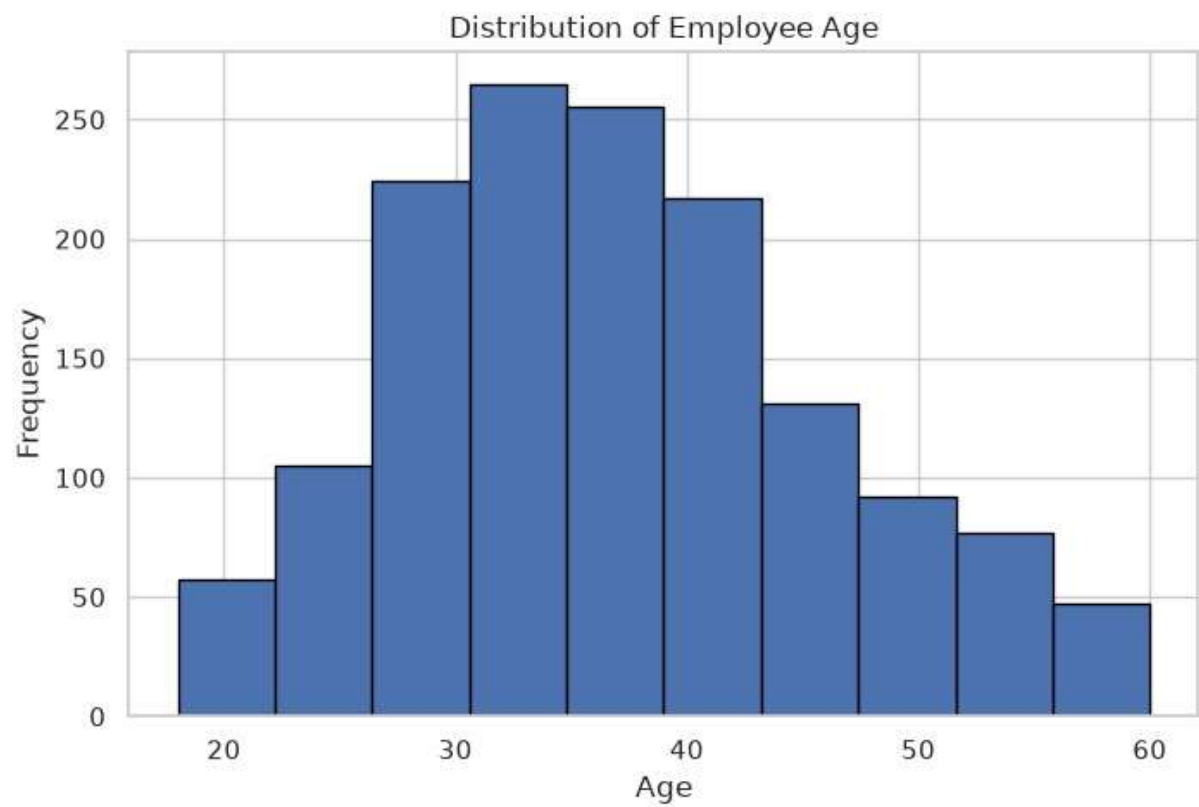
plt.show()
```



```
plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=10, edgecolor="black")

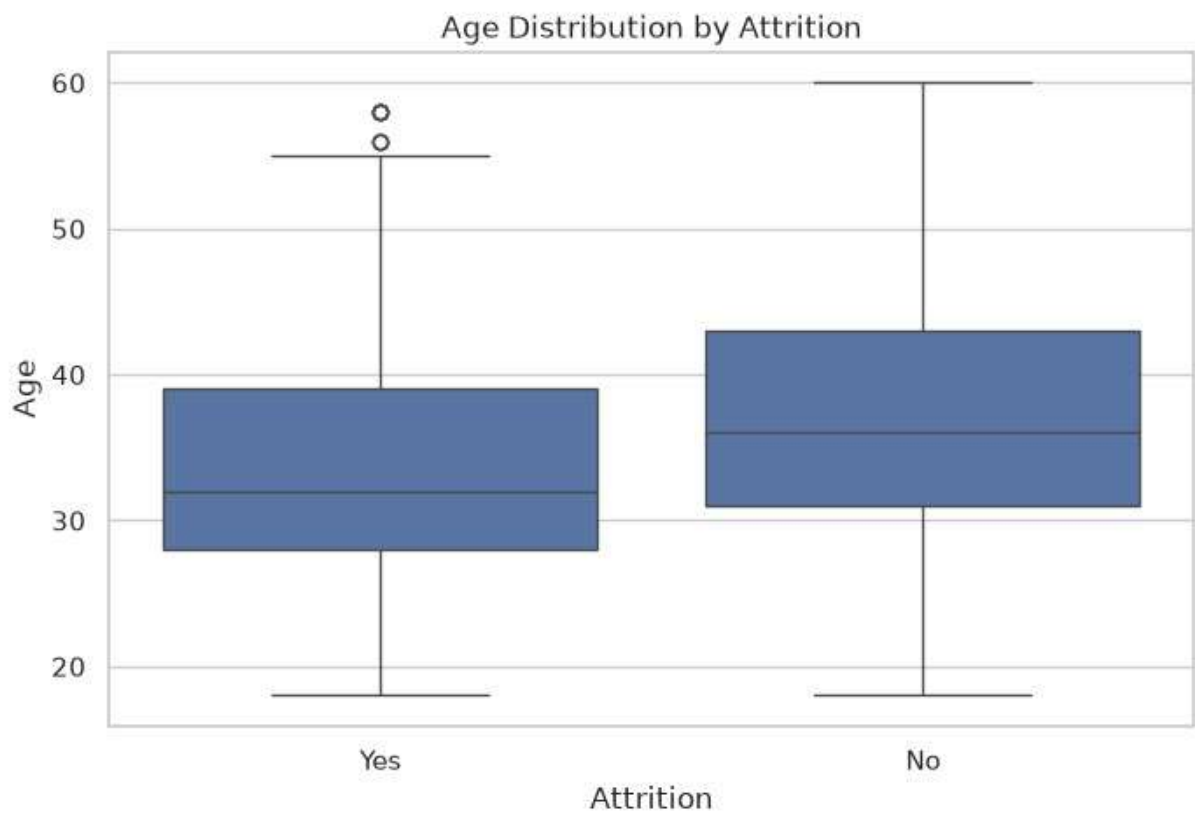
plt.title("Distribution of Employee Age")
plt.xlabel("Age") plt.ylabel("Frequency")
plt.show()
```



```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Attrition", y="Age")

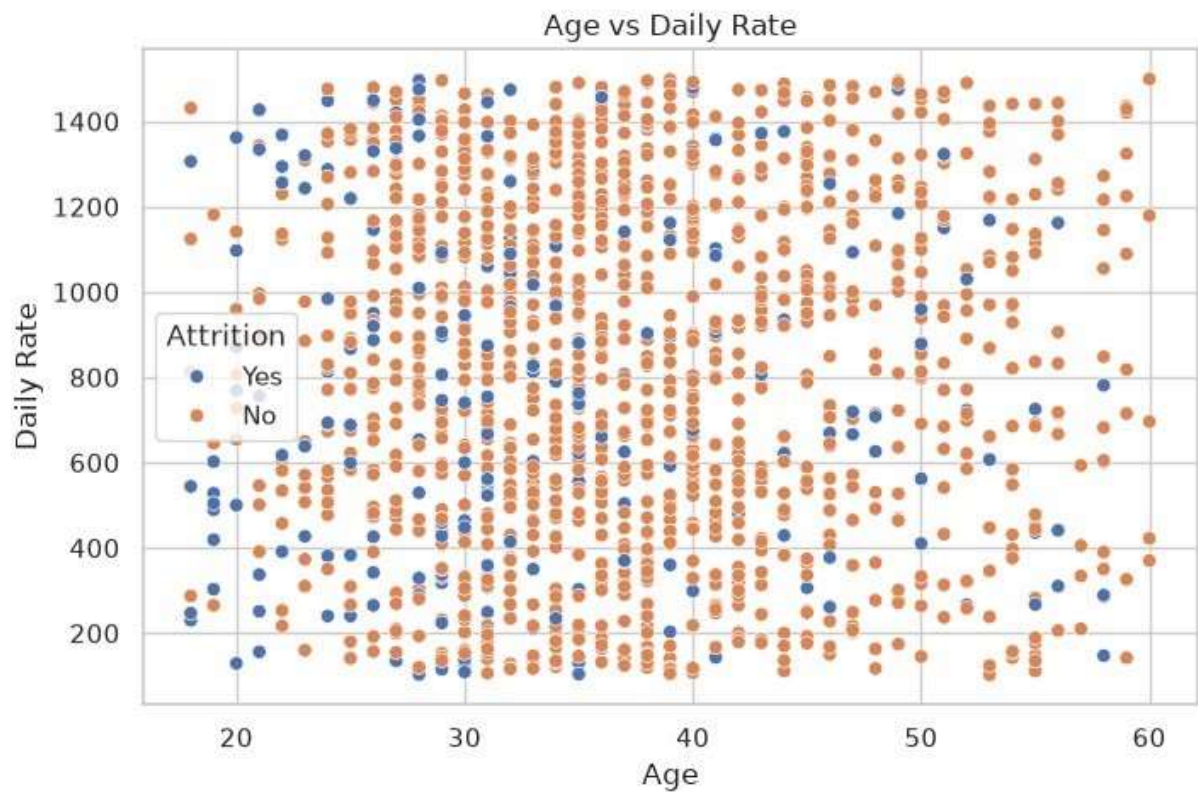
plt.title("Age Distribution by Attrition")
plt.xlabel("Attrition") plt.ylabel("Age")
plt.show()
```



```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Age",
    y="DailyRate",
    hue="Attrition"
)

plt.title("Age vs Daily Rate")
plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.show()
```

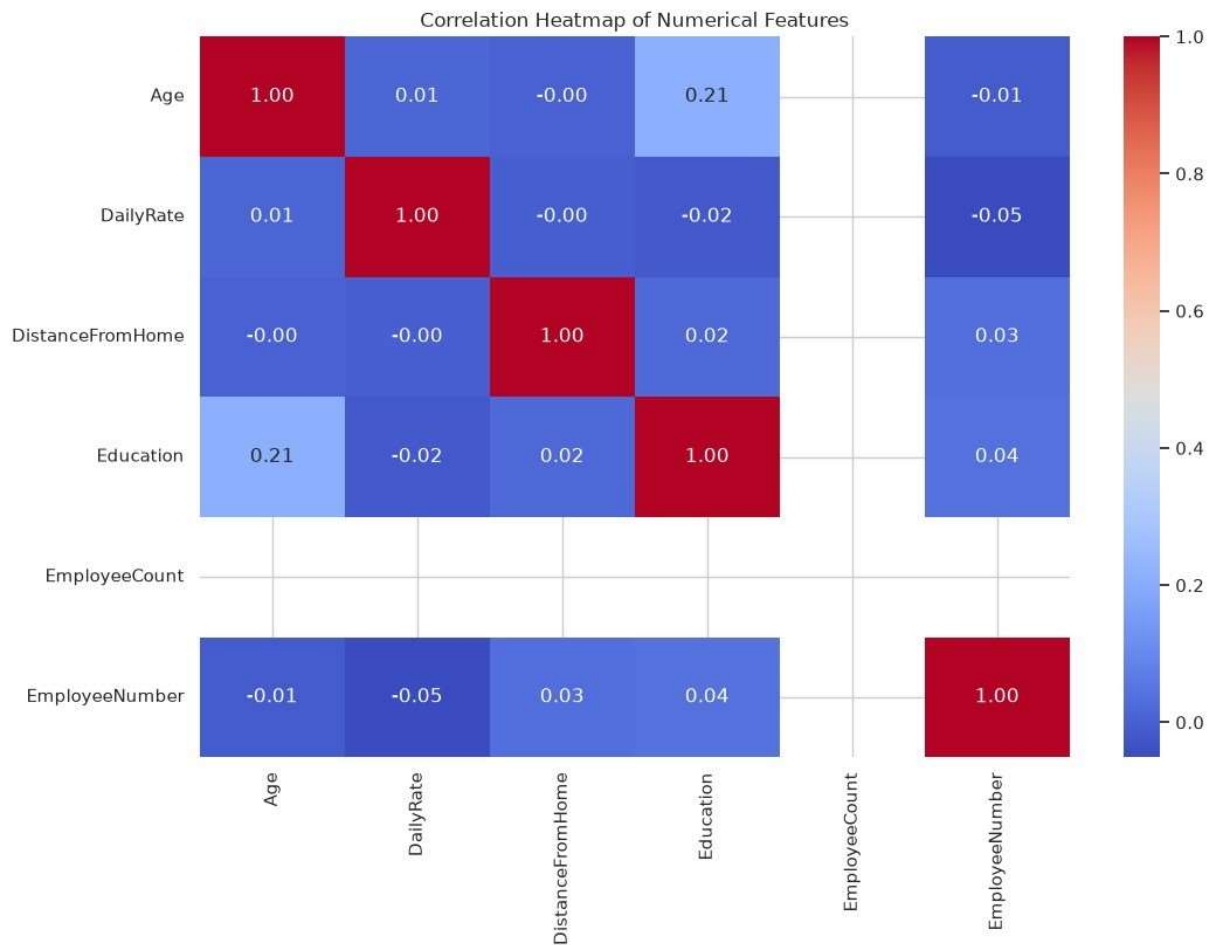


```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(12, 8))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```

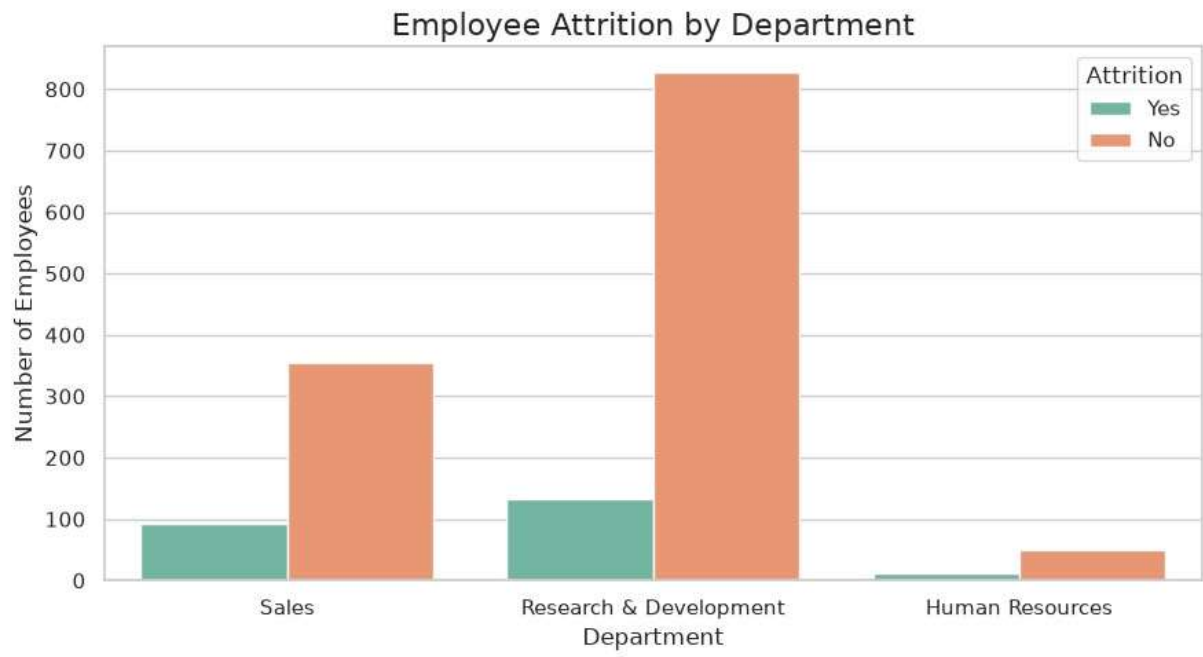


```
plt.figure(figsize=(9, 5))

sns.countplot(
    data=df,
    x="Department",
    hue="Attrition",
    palette="Set2"
)

plt.title("Employee Attrition by Department", fontsize=16)
plt.xlabel("Department", fontsize=12) plt.ylabel("Number
of Employees", fontsize=12) plt.legend(title="Attrition")

plt.tight_layout()
plt.show()
```



```
print("Total Employees:", len(df))

print("\nDepartment-wise Employees:")
print(df["Department"].value_counts())

print("\nAttrition Distribution:")
print(df["Attrition"].value_counts())
print("\nAverage Age:", df["Age"].mean())

print("\nAverage Daily Rate:", df["DailyRate"].mean())
```

Total Employees: 1470

Department-wise Employees:

Department

Research & Development	961
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Sales	446
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Human Resources	63
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Name: count, dtype: int64

Attrition Distribution:

Attrition

No	1233
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Yes	237
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Name: count, dtype: int64

Average Age: 36.923809523809524

Average Daily Rate: 802.4857142857143